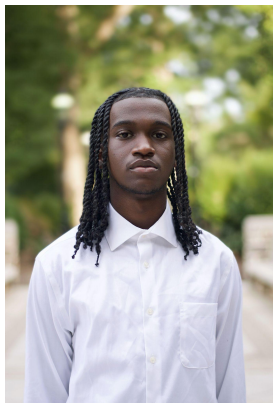


Analyzing Housing Costs, Crime Rates, & School Rating in Chicago

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Meet the Team



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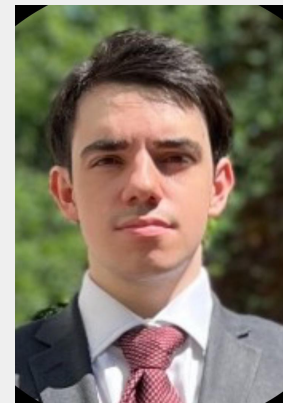
Julian Wilson

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Proposal

How do real estate prices vary based on proximity to amenities such as schools, and Crime rates within Chicago?

Why Chicago:

- Diverse neighborhoods with varying home prices, crime rates, and school quality



Data

- This script loads ZIP code and geographic data, filters for locations in Illinois containing “Chicago,” and saves their ZIP codes with coordinates to a CSV.
- It then loads housing data, filters for Chicago-related cities, keeps only key columns including date-specific ones, renames the ZIP code column, and saves the cleaned data to another CSV. Both steps print the number of records saved.

```
import pandas as pd
import os

column_names = ['Country', 'ZipCode', 'City', 'State', 'StateAbbr', 'County', 'CountyCode',
                'Empty1', 'Empty2', 'Latitude', 'Longitude', 'Accuracy']

df = pd.read_csv('input_data1/Zip Code to GeoID.txt', sep='\t', header=None, names=column_names)

chicago_df = df[(df['StateAbbr'] == 'IL') &
                 (df['City'].str.contains('Chicago', case=False, na=False))][['ZipCode', 'Longitude', 'Latitude']]

chicago_df.to_csv('clean_data2/chicago_il_zipcodes.csv', index=False)

print(f"Chicago ZIP codes file created: {chicago_df.shape[0]} records")
```

```
housing_data = pd.read_csv('input_data1/Zip_zhvi_uc_sfrcondo_tier_0.33_0.67_sm_sa_month.csv')

filtered_housing = housing_data[housing_data['City'].isin(['Chicago', 'Chicago Heights', 'Chicago Ridge'])]

date_columns = [col for col in housing_data.columns if col.startswith('20')]

columns_to_keep = ['RegionName', 'SizeRank', 'State'] + date_columns
filtered_housing = filtered_housing[columns_to_keep]

filtered_housing = filtered_housing.rename(columns={'RegionName': 'Zip Code'})

filtered_housing.to_csv('clean_data2/filtered_chicago_housing.csv', index=False)

print(f"Filtered housing data saved: {filtered_housing.shape[0]} records")
```



Data

- This script filters the Chicago crime dataset to include columns like date, year, crime type, location description, arrest status, and coordinates, then saves the cleaned data to `filtered_chicago_crimes.csv` and prints the number of records.
- It then loads the 2022–2023 Chicago Public Schools dataset, selects key fields related to school name, category, student demographics, performance ratings, and location, adds a “Year” column set to 2023, and saves the result to `cleaned_chicago_schools.csv`.

```
# Select only the required columns from the crime dataset
columns_to_keep = ['Date', 'Year', 'Primary Type', 'Location Description', 'Arrest', 'Latitude', 'Longitude']
cleaned_crime_df = crime_df[columns_to_keep]

# Save the cleaned dataset to a CSV file
cleaned_crime_df.to_csv('clean_data2/filtered_chicago_crimes.csv', index=False)

# Print information about the cleaned dataset
print(f"Cleaned crime data saved: {cleaned_crime_df.shape[0]} records")
```

```
file_path = 'input_data1/Chicago_Public_SchoolsSY2223.csv'
schools_df = pd.read_csv(file_path)

columns_to_keep = [
    'Long_Name', 'Primary_Category', 'Zip', 'Student_Count_Total',
    'Student_Count_Low_Income', 'Student_Count_Special_Ed', 'Student_Count_English_Learners',
    'Student_Count_Black', 'Student_Count_Hispanic', 'Student_Count_White',
    'Student_Count_Asian', 'Student_Count_Native_American', 'Student_Count_Other_Ethnicity',
    'Student_Count_Asian_Pacific_Islander', 'Student_Count_Multi',
    'Student_Count_Hawaiian_Pacific_Islander', 'Student_Count_Ethnicity_Not_Available',
    'Overall_Rating', 'Rating_Status', 'School_Latitude', 'School_Longitude', 'Location'
]

cleaned_schools_df = schools_df[columns_to_keep]

cleaned_schools_df.loc[:, 'Year'] = 2023

cleaned_schools_df.to_csv('clean_data2/cleaned_chicago_schools.csv', index=False)
```

Data

- This script begins by loading four cleaned datasets: ZIP code coordinates, housing data, crime records, and public school information. It prints unique ZIP code counts from each dataset and identifies how many ZIP codes overlap between the ZIP code and housing data. It then ensures all ZIP code fields are consistently formatted as strings for accurate merging.
- The first merge joins ZIP code data with housing data using ZIP codes. Next, it filters the crime data to include only 2023 incidents, rounds coordinates to four decimal places for approximate location matching, and aggregates crime counts by those coordinates. This summarized crime data is then merged with the existing dataset.
- Finally, the school data is merged in based on ZIP codes. The script ensures missing crime counts are filled with zeros, creates a directory for output if it doesn't exist, and saves the fully merged dataset as **final_merged_dataset.csv**, reporting the number of rows and columns in the result.

```
# Load all datasets
zip_code_df = pd.read_csv('clean_data2/chicago_il_zipcodes.csv')
housing_df = pd.read_csv('clean_data2/filtered_chicago_housing.csv')
crime_df = pd.read_csv('clean_data2/filtered_chicago_crimes.csv')
schools_df = pd.read_csv('clean_data2/cleaned_chicago_schools.csv')

# Print unique counts to understand the data
print(f"ZIP codes unique count: {zip_code_df['ZipCode'].nunique()}")
print(f"Housing ZIP codes unique count: {housing_df['Zip Code'].nunique()}")
print(f"Schools ZIP codes unique count: {schools_df['Zip'].nunique()}")

# Check for common values in the joining columns
print(f"Common ZIP codes between zip_code_df and housing_df: {len(set(zip_code_df['ZipCode']).astype(str)) & set(housing_df['Zip Code'].astype(str))}")

# Look at data types to ensure they match
print(f"ZIP code types - zip_code_df: {zip_code_df['ZipCode'].dtype}, housing_df: {housing_df['Zip Code'].dtype}, schools_df: {schools_df['Zip'].dtype}")

# Convert ZIP codes to strings for consistent matching
zip_code_df['ZipCode'] = zip_code_df['ZipCode'].astype(str)
housing_df['Zip Code'] = housing_df['Zip Code'].astype(str)
schools_df['Zip'] = schools_df['Zip'].astype(str)

# First merge: ZIP codes with housing data
merged_df = pd.merge(
    zip_code_df,
    housing_df,
    left_on='ZipCode',
    right_on='Zip Code',
    how='inner'
)
print(f"After first merge: {len(merged_df)} rows")

# For crime data, check if exact coordinates match exists
# If not, consider using approximate matching
# First, create a copy of the filtered data to avoid the SettingWithCopyWarning
crime_df_2023 = crime_df[crime_df['Year'] == 2023].copy()

# Round coordinates for approximate matching
merged_df['Latitude_rounded'] = merged_df['Latitude'].round(4)
merged_df['Longitude_rounded'] = merged_df['Longitude'].round(4)
crime_df_2023.loc[:, 'Latitude_rounded'] = crime_df_2023['Latitude'].round(4)
crime_df_2023.loc[:, 'Longitude_rounded'] = crime_df_2023['Longitude'].round(4)

# Group crime data by rounded coordinates and count incidents
crime_summary = crime_df_2023.groupby(['Latitude_rounded', 'Longitude_rounded']).size().reset_index(names='crime_count')

# Second merge: Add crime summary data using rounded coordinates
merged_df = pd.merge(
    merged_df,
    crime_summary,
    on=['Latitude_rounded', 'Longitude_rounded'],
    how='left'
)
print(f"After second merge: {len(merged_df)} rows")

# Fill NaN values in crime_count
merged_df['crime_count'] = merged_df['crime_count'].fillna(0)

# Third merge: Add schools data
merged_df = pd.merge(
    merged_df,
    schools_df,
    left_on='ZipCode',
    right_on='Zip',
    how='left'
)
print(f"After third merge: {len(merged_df)} rows")

# Create output directory if it doesn't exist
output_dir = 'merged_output'
if not os.path.exists(output_dir):
    os.makedirs(output_dir)

# Save the final merged dataset
merged_df.to_csv(os.path.join(output_dir, 'final_merged_dataset.csv'), index=False)
print(f"Merged dataset saved with {len(merged_df)} rows and {len(merged_df.columns)} columns")
```

Data Preparation

1. Load & Filter Dataset

The script loads the `final_merged_dataset.csv`, which contains ZIP codes, housing prices, crime statistics, and school data. It keeps only the 2023 housing price columns (e.g., `2023-01-31`), calculates their yearly average, and drops the individual monthly columns.

2. Column Cleanup

Unnecessary columns such as `Year`, `State`, duplicate coordinates, and `Location` are removed. The result is a streamlined dataset that includes the average 2023 housing cost per ZIP code, crime counts by type and total, and school information including names, demographics, and ratings.

3. School Data Focus

A focused subset called `df1` is created, containing key columns: `ZipCode`, `Total_Crimes`, `Student_Count_Total`, `Student_Count_Low_Income`, `Overall_Rating`, `Rating_Status`, `average_housing_cost_2023`, and `Long_Name`. Rows with missing values (e.g., ZIP codes without schools) are dropped.

4. Categorical to Numerical Transformations

School ratings are converted from text to numbers: Level 1 or 1+ is mapped to 1 (best), Level 2 or 2+ to 2, and Level 3 to 3. Schools rated as “Inability to Rate” are treated as missing values and excluded. Similarly, rating status is transformed numerically: “GOOD STANDING” becomes 3, “PROVISIONAL SUPPORT” becomes 2, and “INTENSIVE SUPPORT” becomes 1. “NOT APPLICABLE” entries are excluded as well.

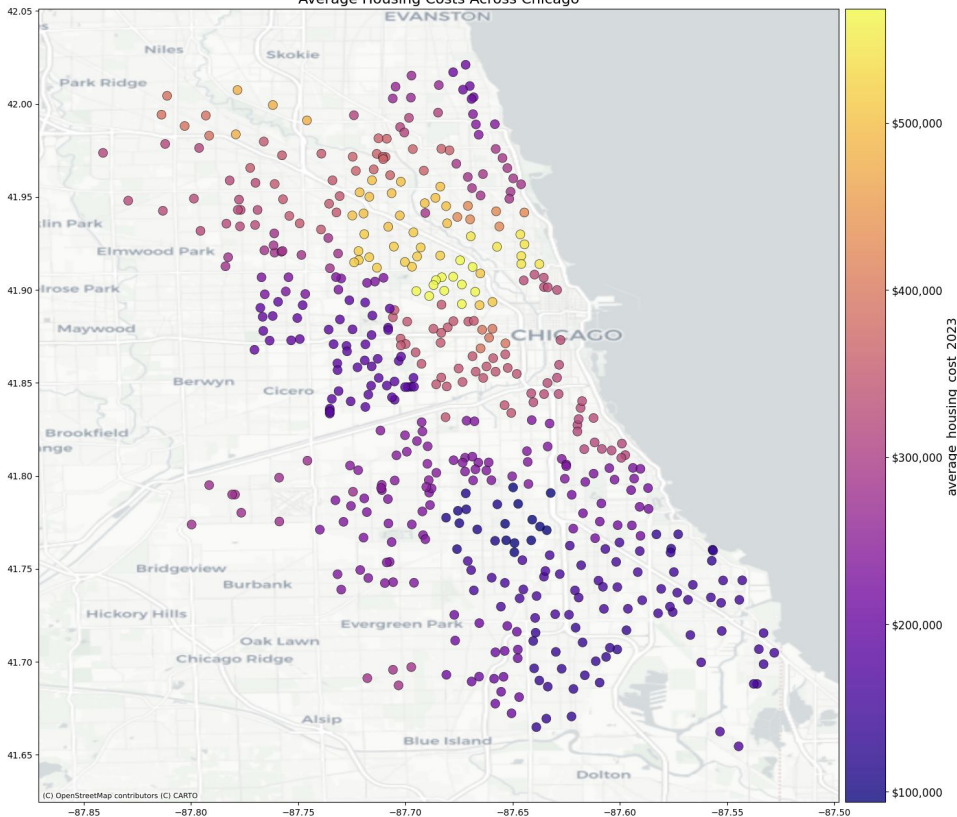
5. Final Output

The resulting cleaned DataFrame (`df1`) is ready for analysis and allows for the exploration of the relationships between crime rates, school quality, and housing costs by ZIP code.



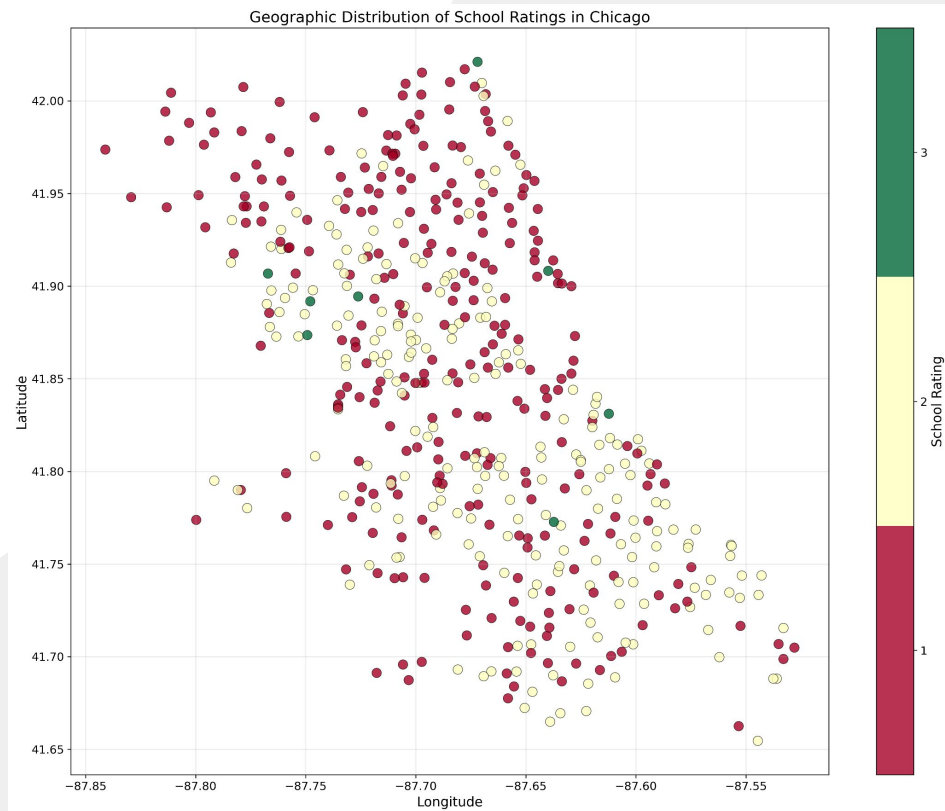
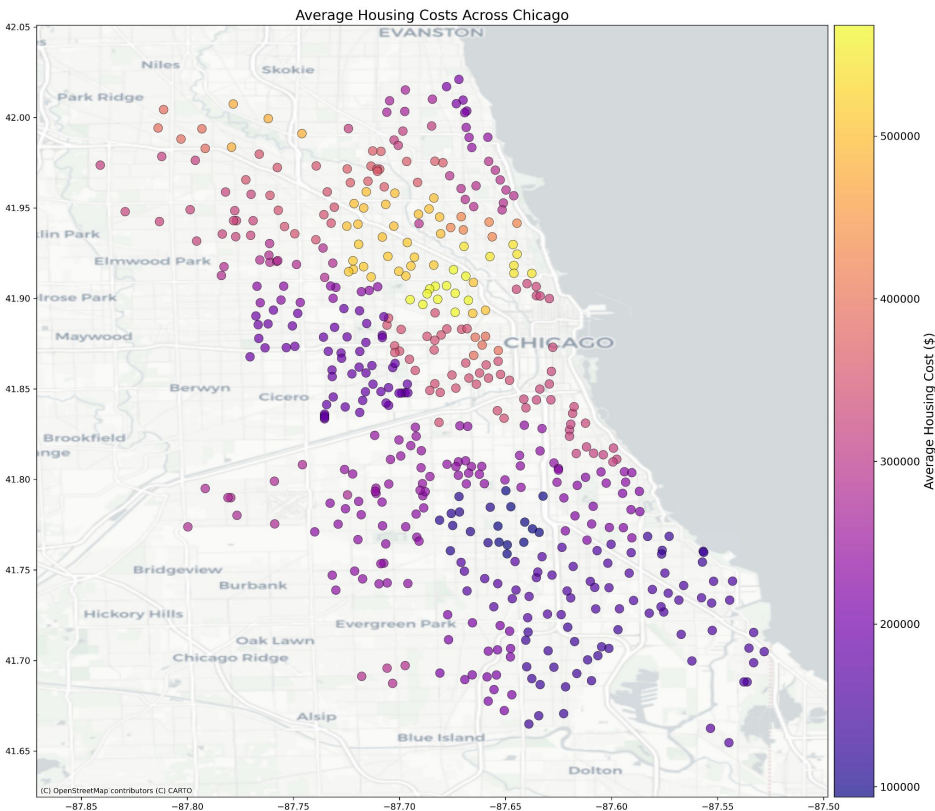
Distribution of Housing Prices

Average Housing Costs Across Chicago



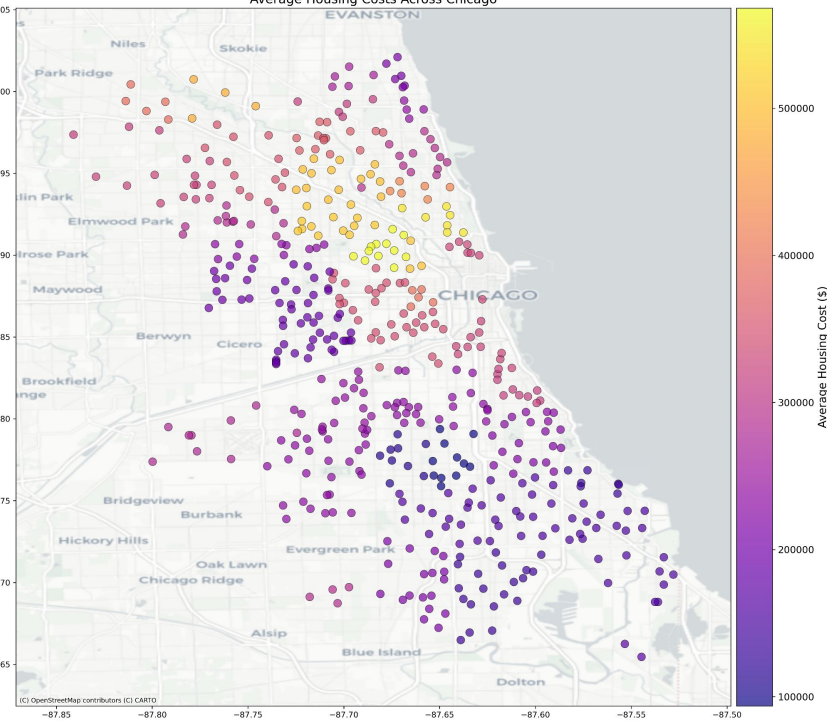
- Higher Average Home Costs: Downtown & North Side
- Lower Average Home Costs: South/West of city center

Housing Costs + School Ratings



Housing Costs + Crime

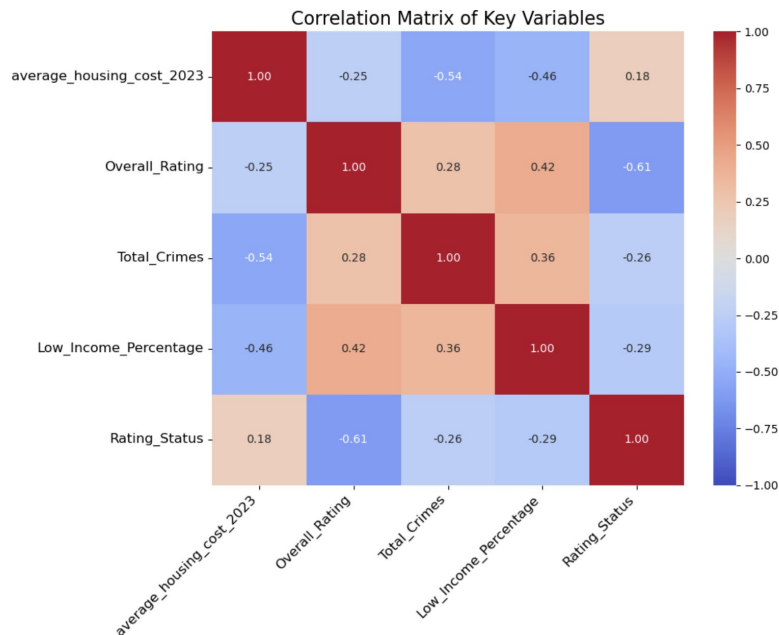
Average Housing Costs Across Chicago



Geographic Distribution of Crime Rates in Chicago



Key Takeaways



- Higher home cost reflect neighborhood quality
- School rating and crime are interlinked
- There is a strong correlation between home costs, school quality, and crime throughout Chicago

Thank You!



Your Questions, Our Answers



References

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<https://www.census.gov/programs-surveys/geography/guidance/geo-identifiers.html>

<https://www.fhfa.gov/data/hpi/datasets?tab=hpi-datasets>

<https://data.cityofchicago.org/browse?q=property+prices&sortBy=relevance&pageSize=20&page=1>

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